
BRAVO COMPANY
COMMUNICATIONS DOMINATE WARFARE

DATA SYSTEMS ADMINISTRATOR COURSE

DS.02.01

OPERATING SYSTEMS

INTRODUCTION TO OPERATING SYSTEMS

OVERVIEW

- Characteristics of Operating Systems
- Types of Operating Systems
- Servers
- Types of Servers
- Operating System Installation Types
- Server Operating Systems Edition Types

LEARNING OBJECTIVES

- Refer to your student handout for detailed list of your Terminal and Enabling learning objectives.

CHARACTERISTICS OF OPERATING SYSTEMS

Characteristics of operating systems

- ***What is an Operating System (OS)?***
  - A type of software and it works as an interface between the user and server to perform all the tasks necessary.
  - Every device works with an operating system (OS) to run all the programs & applications.
  - Without an operating system, a computer or server is useless.

Characteristics of operating systems

- ***Components of an Operating System (OS)***
  1. ***Kernel*** - Provides the basic level of control on all the computer peripherals
  2. ***Process Execution*** - The program can connect through the hardware device by simply following procedures & principles configured into the OS
  3. ***Interrupt*** - They give a reliable technique for the OS to communicate & react to their surroundings.
  4. ***Memory Management*** - Memory management work can be divided into three important groups like memory management of hardware, OS and application memory management.

Characteristics of operating systems

- ***Components of an Operating System (OS)***

  5. Multitasking - Multitasking in an OS allows an operator to execute one or more computer tasks at a time

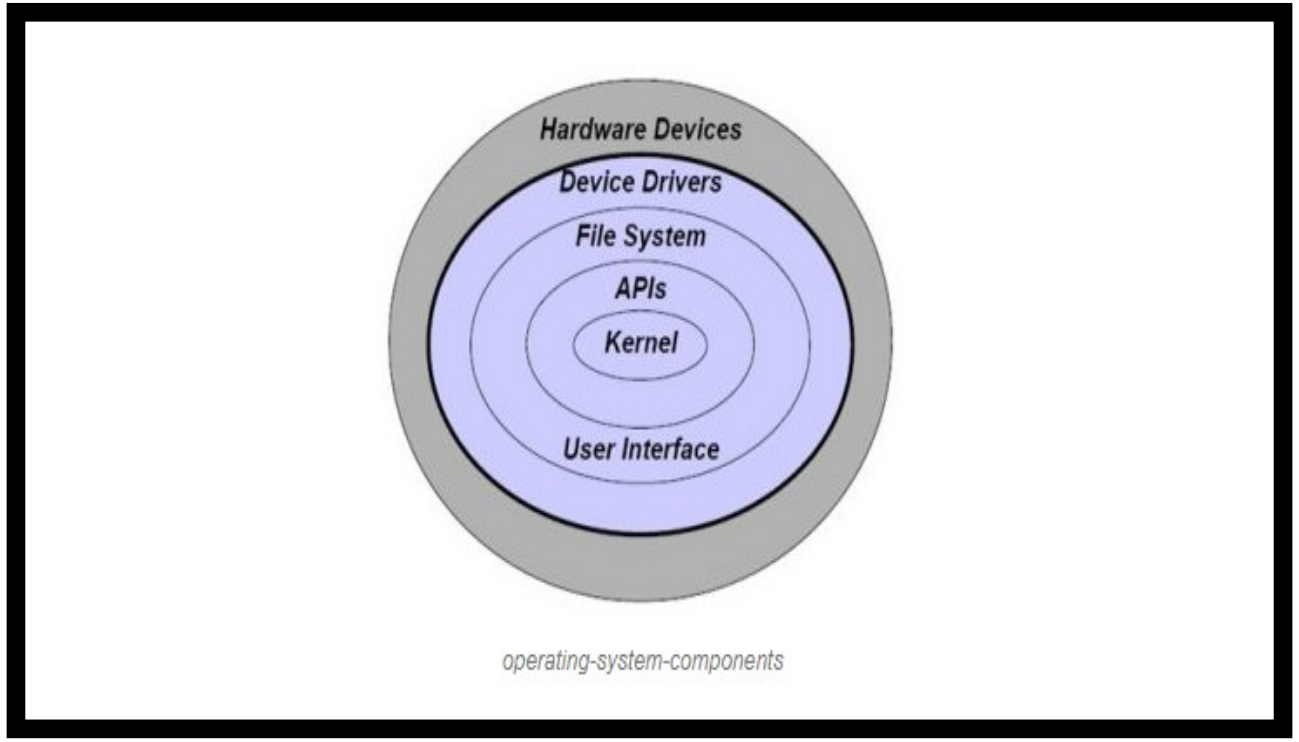
  6. Networking - When the processor interacts with each other through communication lines

  7. Security - This will allow monitoring the request from accessibility to other resources

  8. User Interface - A GUI or user interface (UI) is the part of an OS that permits an operator to get the information

Characteristics of operating systems

- ***Components of an Operating System (OS)***

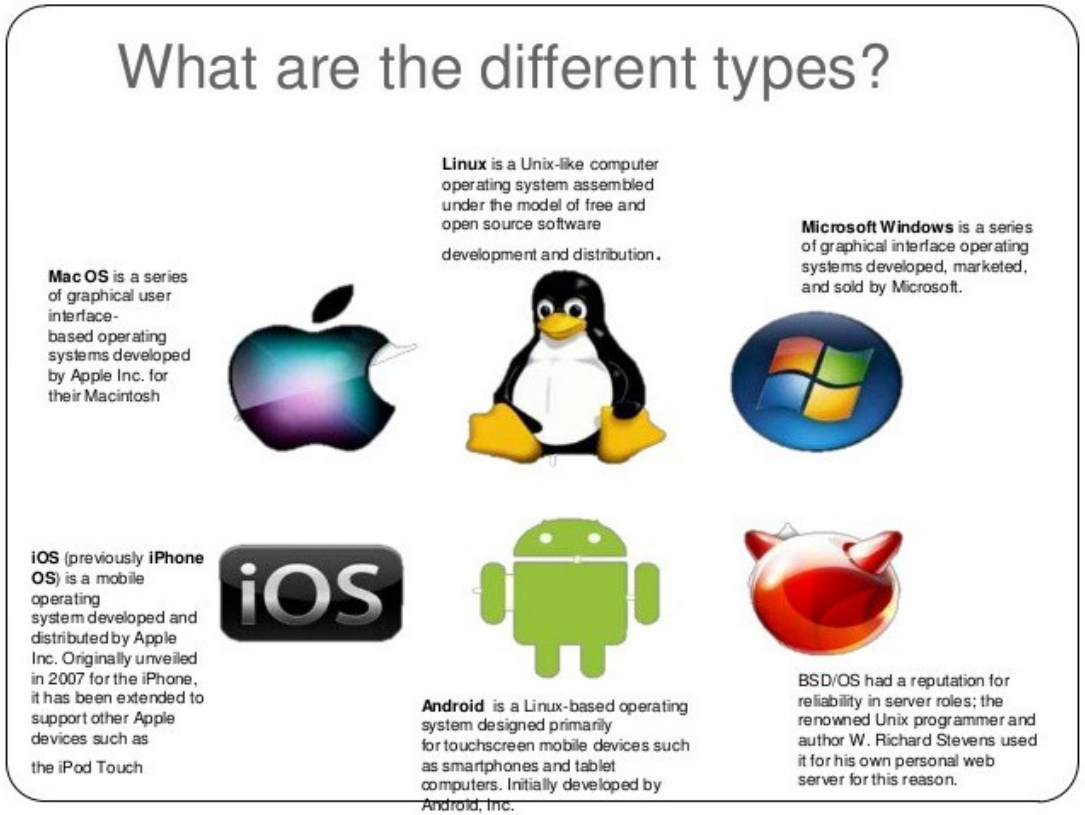


*The main components of an OS mainly include kernel, API or application program interface, user interface & file system, hardware devices and device drivers.

TYPES OF OPERATING SYSTEMS

Types of operating systems

- These five OS types are likely what run any data center that can control servers, phones, computers, or other mobile devices like a



Types of Operating Systems

Types of operating systems

1. *Microsoft Windows*

- Microsoft Windows is one of the popular operating system types and is preloaded on most new PC hardware.
- Many users prefer Microsoft Windows because they say it's compatible with many other kinds of software.
- Many kinds of computer programs run best on Microsoft Windows because they're developed by Microsoft.
- Most USMC networks/domains include Microsoft Windows Servers

Types of operating systems

2. *Apple macOS*

- Head-to-head in the competition with Microsoft Windows is Apple's macOS.
- Apple and Macintosh computers run on the proprietary macOS and OS X system, the first of which launched 20 years ago
- Apple macOS has not made its way onto USMC networks yet however, it could be within the future.



Types of operating systems

3. *Google's Android OS*

- Companies including Google use to run its Android mobile smartphones and tablets
- Is based on Linux distribution and other open source software.
- Android OS is primarily used within the ***USMC*** Mobility Infrastructure.
- Marines that are transferred to “Base G6’s” are required to provide support for their end users.

Types of operating systems

4. Apple iOS

- Mobile operating system used exclusively for iPhones
- Many users appreciate the unique user interface with touch gestures, and the ease of use that iOS offers
- iPhones are now operating on the Marine Corps Enterprise Network and fall under the Mobility Infrastructure.

Types of operating systems

5. *Linux Operating System*

- Different from Windows and Apple in that it’s not a proprietary software.
- Anyone can modify and distribute it.
- Linux is popular because of its ease of customization and offers a variety of options to those who understand how to use it.
- Linux systems that are used on Marine Corps networks are as follows;
  - Cisco Call Manager
  - ACAS
  - Even some Remote Administration Servers

SERVERS

SERVERS

- **Definition of a Server**
  - A server is a computer program or device that provides a service to another computer program and its user, also known as the client.
  - They are typically virtualized (taught in Annex A&B)
  - Clients can be other users making request or other servers on the network
  - A server may be designed to do a single task or multiple task
  - 0671’s have to configure and administer many different types of servers

SERVERS

- **How a Server Works**
  - Must be configured to listen to requests from clients on a network connection
  - Will often perform numerous additional tasks as part of a single request and response, including verifying the identity



SERVERS

- **Types of Servers**
  1. **Application Servers**
     - Different applications can run irrespective of what they are or what functions they will perform
     - Utilized in developing, processing and running web-based applications

  2. **Cloud Servers**
     - Highly distributed, massive storage and high-end processing power servers. Utilized in developing, processing and running web-based applications
     - Mainly used by “Cloud Computing” to serve multiple request at one time

SERVERS

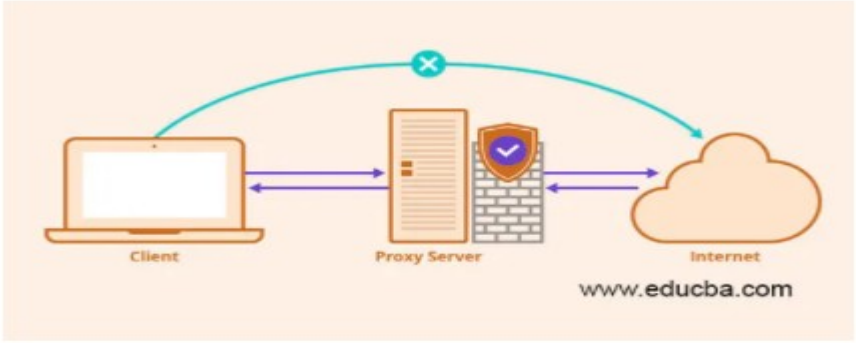
- **Types of Servers**
  3. **Database Servers**
     - The bigger organizations utilize a database for storage and users can access the data by executing a query using any query language.
     - *SCCM and SQL Servers contain massive db’s that are administered by 0671’s.

  4. **Mail Servers**
     - A centralized or remote computer that holds e-mail messages for the client on a particular network
     - USMC Networks currently use Microsoft Exchange 2016, 2019 and Exchange O365

SERVERS

- **Types of Servers**
  5. **Domain Name Service Server**
     - Receives a request having a domain name hostname and responds back corresponding IP address
     - DNS uses port 53 for providing the service.

  6. **Proxy Servers**
     - It caches all the pages accessed through the network which will allow users who want to visit that same page



OPERATING SYSTEM INSTALLATION TYPES

OPERATING SYSTEM INSTALLATION TYPES

- ***The three most common installation methods of Windows***

1. ***DVD Boot/ISO***
    1. Any file that ends in **.iso** is an exact copy of a disk
    2. ISO copies are referred to as "images" of the original
    3. The ISO is a perfect copy of what would be on an original Windows CD or DVD.

2. ***Distribution Share Installation***
    1. Optional set of folders that contain custom scripts, images, branding, applications, drivers, and other files.
    2. During installation, Windows connects to the path of the server share by using the credentials that you specify in an answer file
    3. Seldom used on USMC Networks (NOT SCCM)

OPERATING SYSTEM INSTALLATION TYPES

- ***The three most common installation methods of Windows***

3. Image Based Installation

1. Leveraging an imaging solution to capture an image based on a sector-by-sector copy of the reference computer.
2. A good example of this is Microsoft's System Center Configuration Manager (SCCM).
    1. There are two types of installation
        1. Guided Install
        2. Manual Install

OPERATING SYSTEM INSTALLATION EDITIONS

OPERATING SYSTEM INSTALLATION EDITIONS

Features available generally	Windows Server 2019 Standard	Windows Server 2019 Datacenter
Best Practices Analyzer	Yes	Yes
Direct Access	Yes	Yes
Dynamic Memory (in virtualization)	Yes	Yes
Hot Add/Replace RAM	Yes	Yes
Microsoft Management Console	Yes	Yes
Minimal Server Interface	Yes	Yes
Network Load Balancing	Yes	Yes
Windows PowerShell	Yes	Yes
Server Core installation option	Yes	Yes
Server Manager	Yes	Yes
SMB Direct and SMB over RDMA	Yes	Yes
Software-defined Networking	No	**Yes**
Storage Migration Service	Yes	Yes
Storage Replica	Yes, (1 partnership and 1 resource group with a single 2TB volume)	Yes, **unlimited**
Storage Spaces	Yes	Yes
Storage Spaces Direct	No	**Yes**
Volume Activation Services	Yes	Yes
VSS (Volume Shadow Copy Service) integration	Yes	Yes
Windows Server Update Services	Yes	Yes
Windows System Resource Manager	Yes	Yes
Server license logging	Yes	Yes
Inherited activation	As guest if hosted on Datacenter	**Can be a host or a guest**
Work Folders	Yes	Yes

** Details from Microsoft website at https://docs.microsoft.com/en-us/windows-server/get-started-19/editions-comparison-19

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